



Inspiring Excellence

# **CSE 251**

# **Electronic Devices and Circuits**

## **Lecture 9**

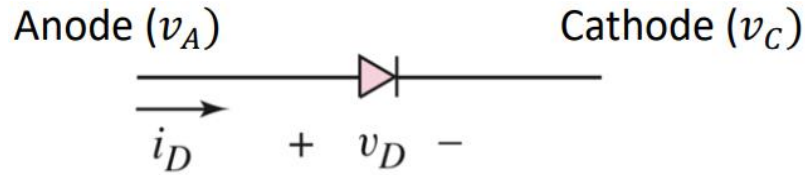
**Course instructor:**

**Mir Hamidul Hussain**

**Lecturer, Department of Computer Science and  
Engineering, School of Data and Sciences, BRAC  
University**

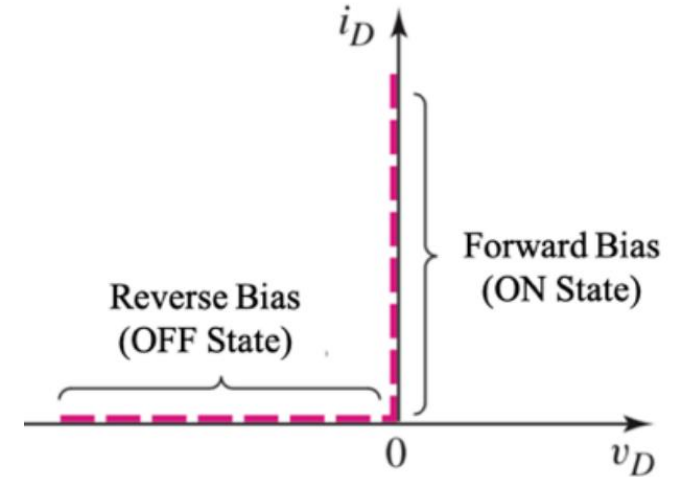
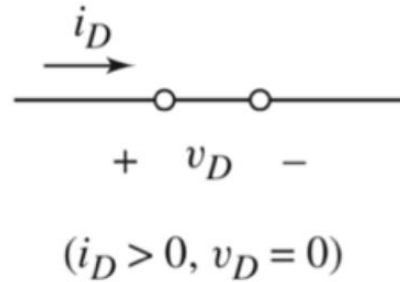
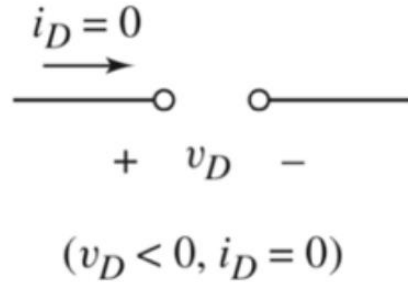
***Email: [hamidul.hussain@bracu.ac.bd](mailto:hamidul.hussain@bracu.ac.bd)***

# Review: Diode Models

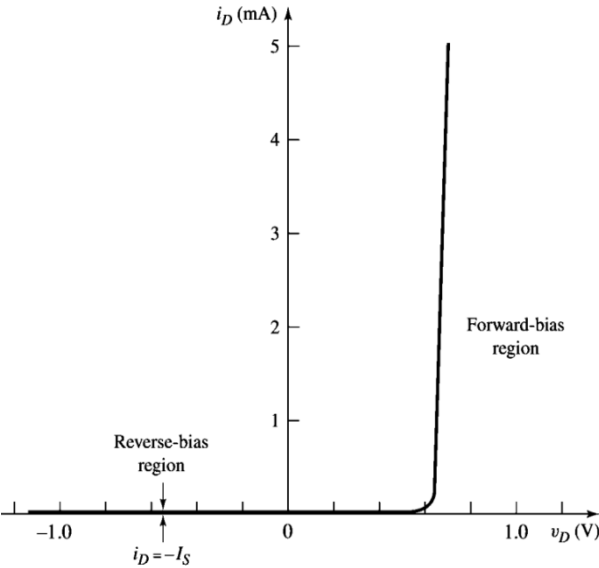


**OFF State: Open circuit**

**ON State: Short circuit**



# Real diode



Relation between diode current and diode voltage:

$$i_D = I_S \left( e^{\frac{v_D}{\eta V_T}} - 1 \right)$$

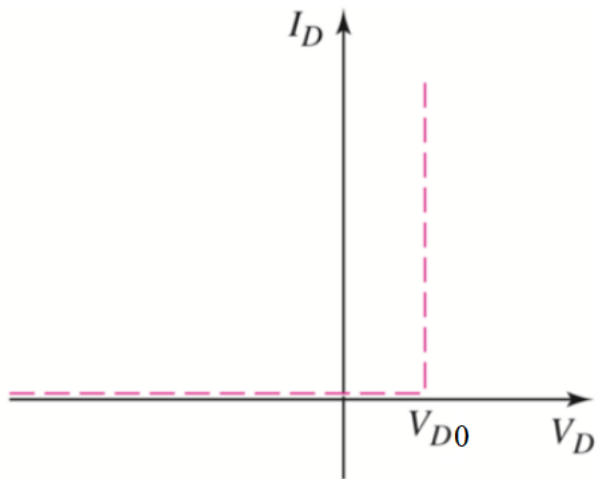
where  $v_D (= v_A - v_C)$  is the voltage across the diode,  $i_D$  is the current through the diode (from anode to cathode) and  $V_T$ , called the thermal voltage, is a temperature dependent constant. For temperature  $T = 300K$ ,  $V_T = 25 \text{ mV}$ .

$\eta$  is called the ideality factor (try to recall, you measured this in the lab!)

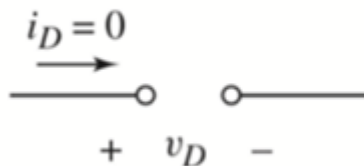
**I-V characteristics of a  
real diode**

# Modeling the real diode

1. Ideal diode model
- 2. Constant voltage drop (CVD) model**
3. CVD+R model

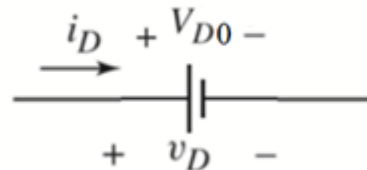


**OFF State: Open circuit**



$$(v_D < V_{D0}, i_D = 0)$$

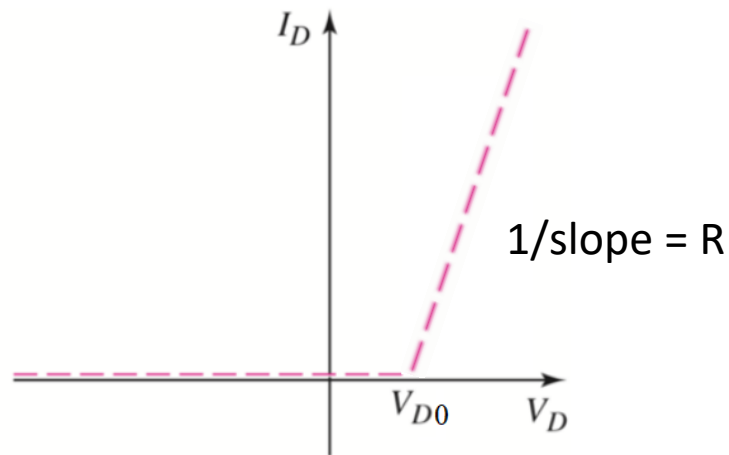
**ON State: Voltage source**



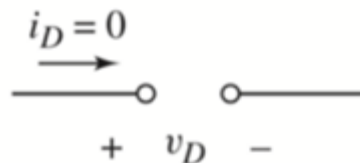
$$(i_D > 0, v_D = V_{D0})$$

# Modeling the real diode

1. Ideal diode model
2. Constant voltage drop (CVD) model
- 3. CVD+R model**

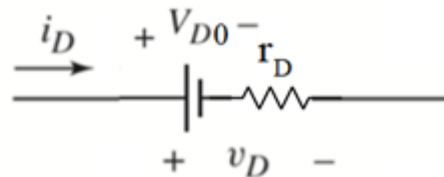


**OFF State: Open circuit**



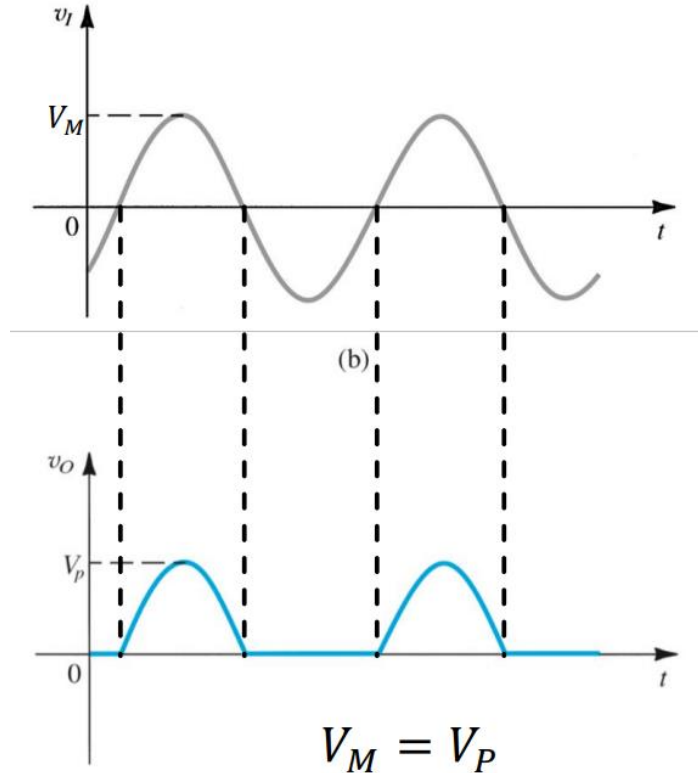
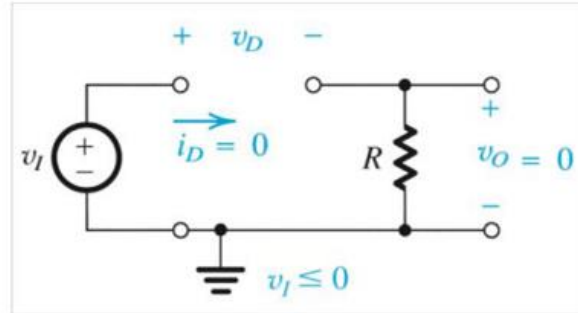
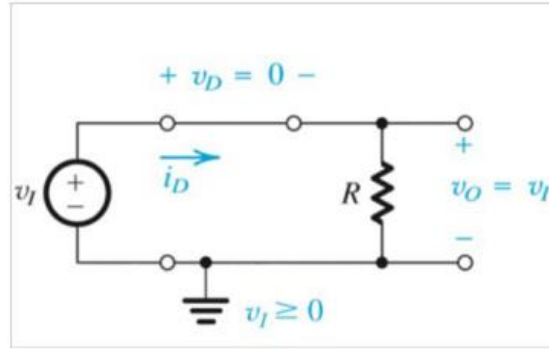
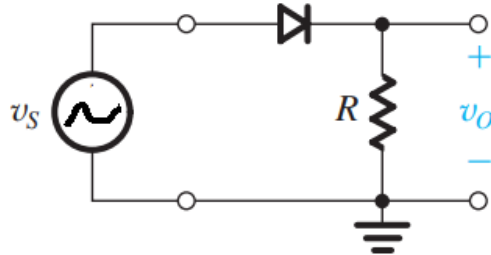
$$(v_D < V_{D0}, i_D = 0)$$

**ON State: Voltage source**

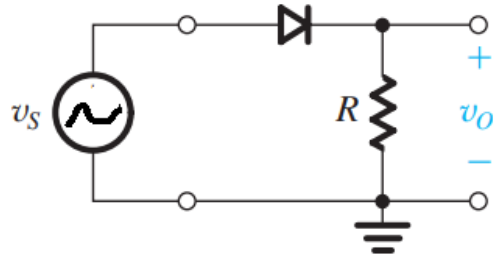


$$(i_D > 0, v_D = V_{D0} + i_D r_D)$$

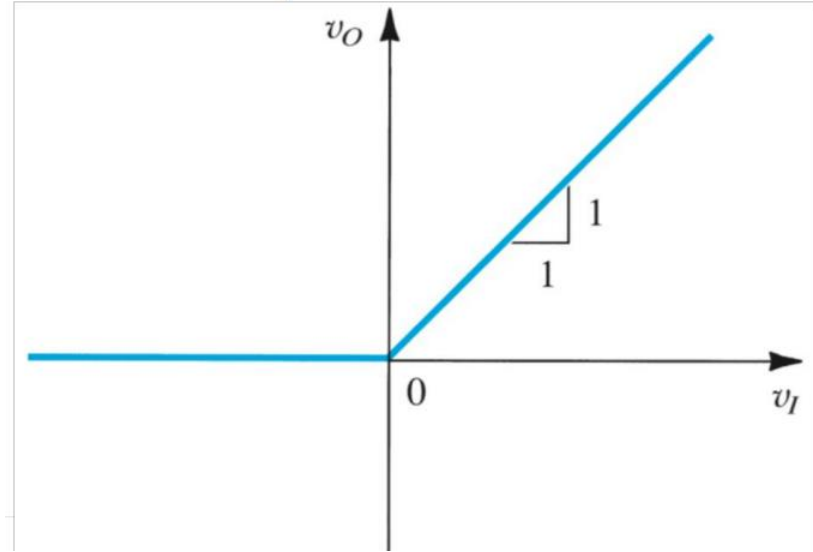
# Half-wave rectifier (ideal diode model)



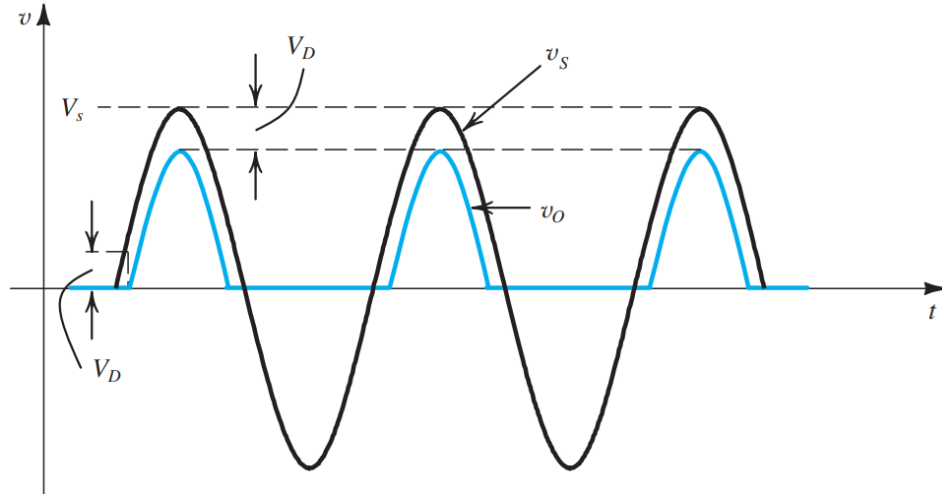
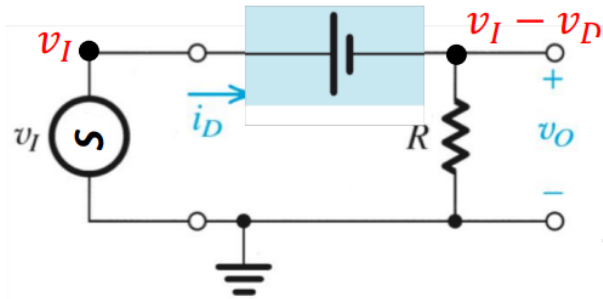
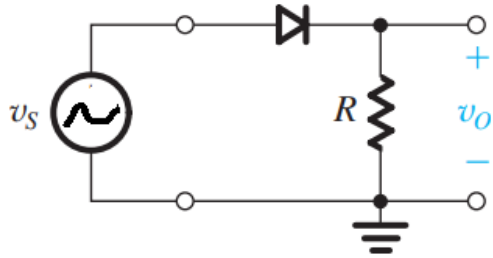
# Half-wave rectifier (ideal diode model)



*Transfer Characteristics*



# Half-wave rectifier (CVD model)



$$v_I = V_M \sin \omega t$$

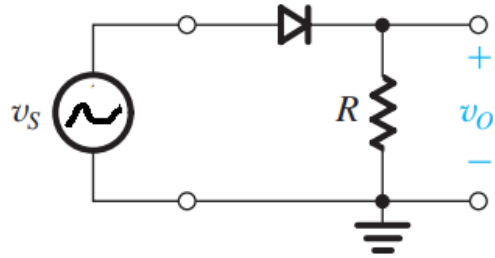
$$v_O = V_M \sin \omega t - V_D$$

$$V_p = \text{peak of output}$$

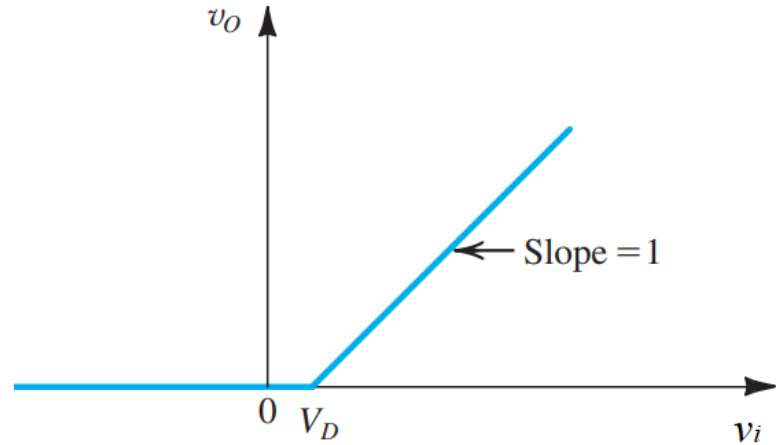
$$= V_M - V_D$$



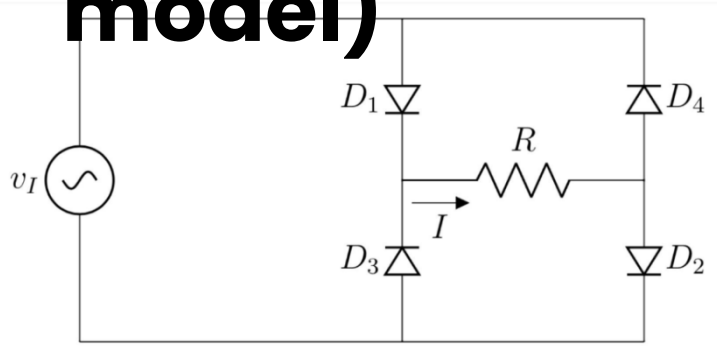
# Half-wave rectifier (CVD model)



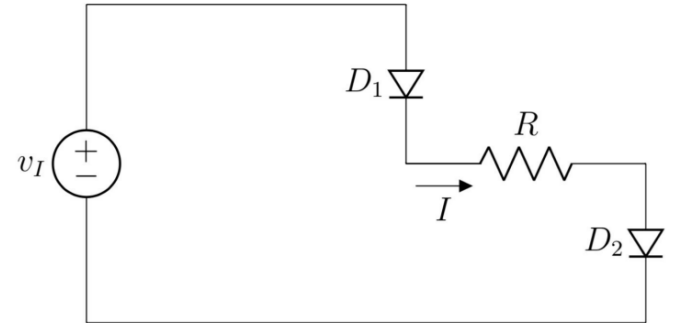
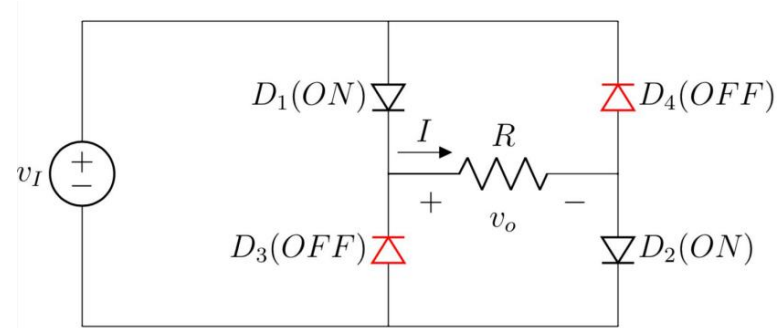
*Transfer Characteristics*



# Full-wave rectifier (ideal diode & CVD model)

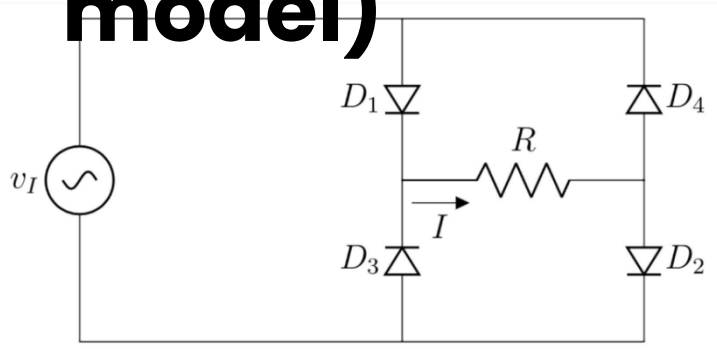


**(+) half-cycle**

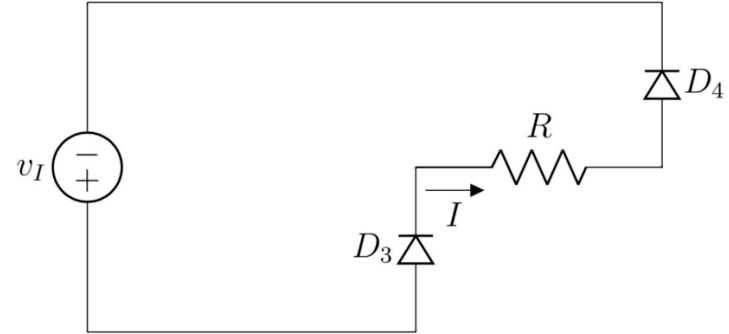
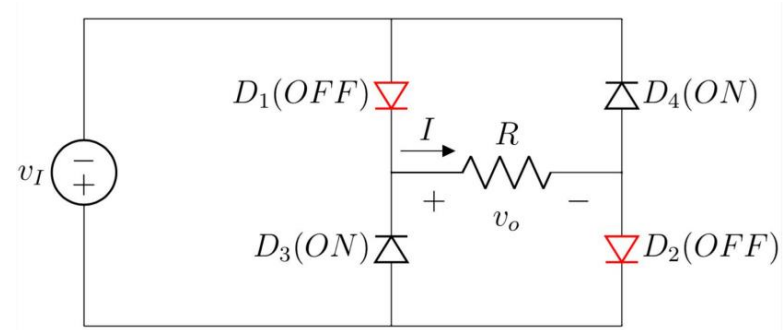


$$v_o = v_I - 2V_D$$

# Full-wave rectifier (ideal diode & CVD model)

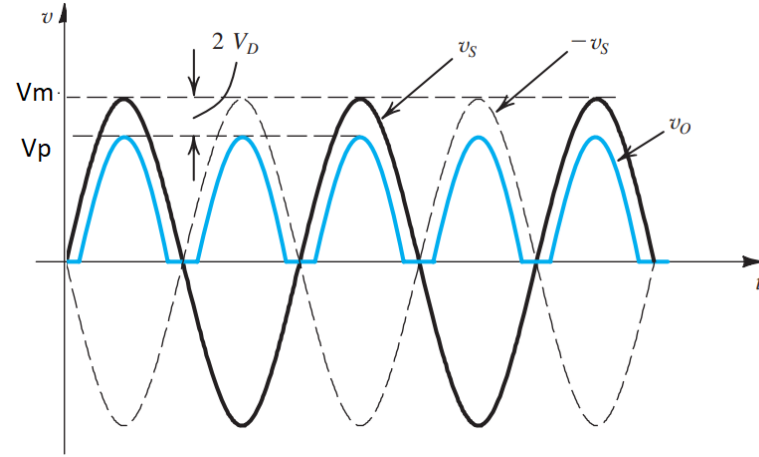
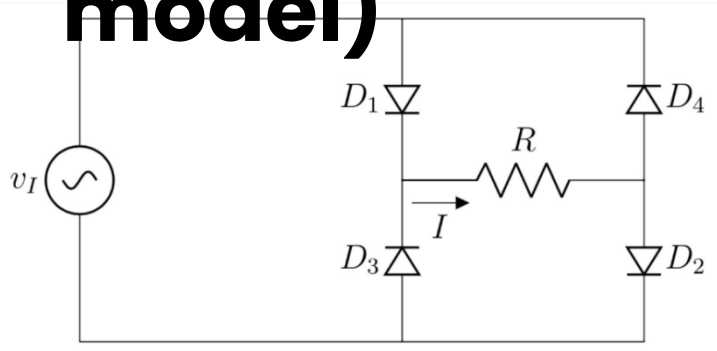


**(-) half-cycle**

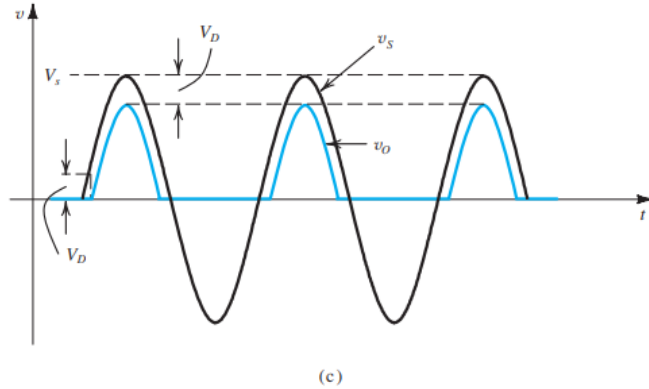
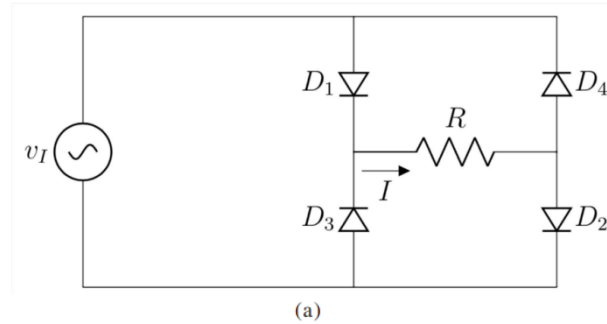
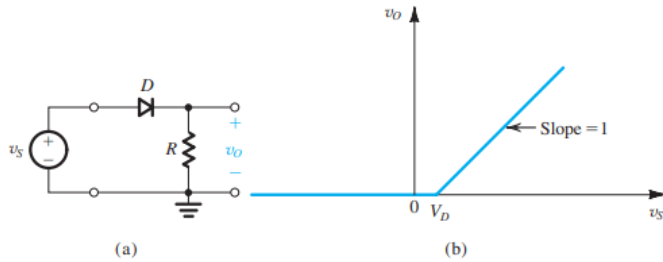


$$v_o = -v_I - 2V_D$$

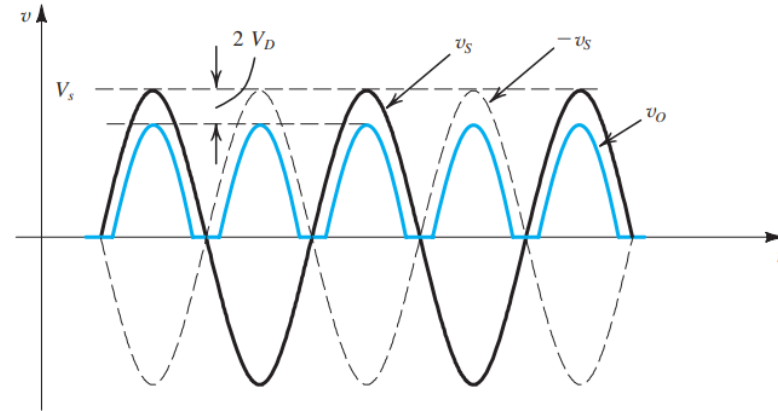
# Full-wave rectifier (ideal diode & CVD model)



# Half-wave and Full-wave rectifier

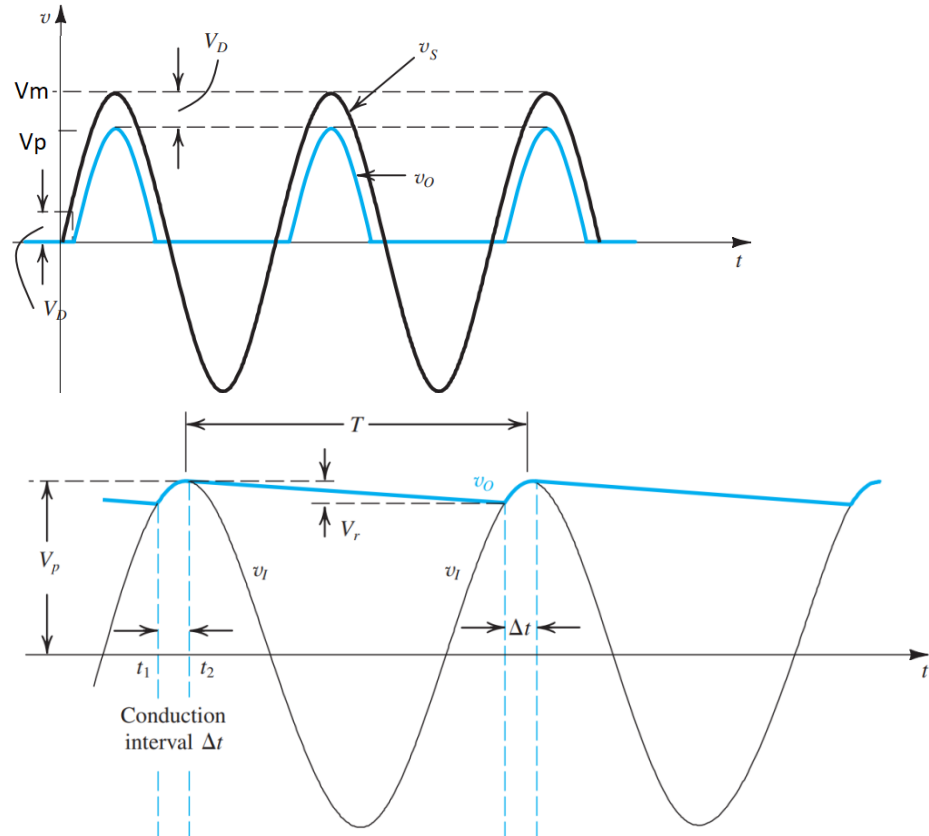
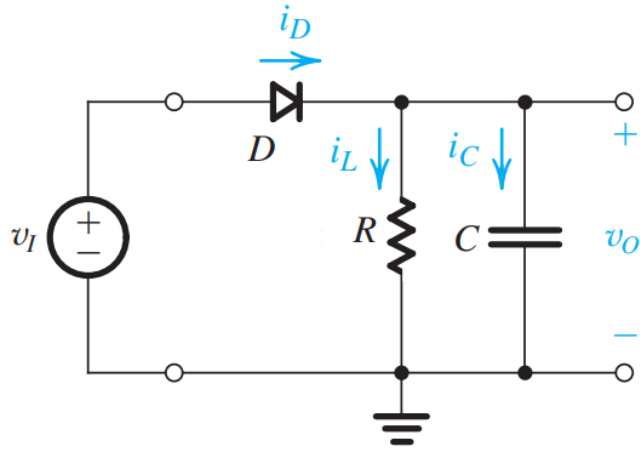


**Half-wave**

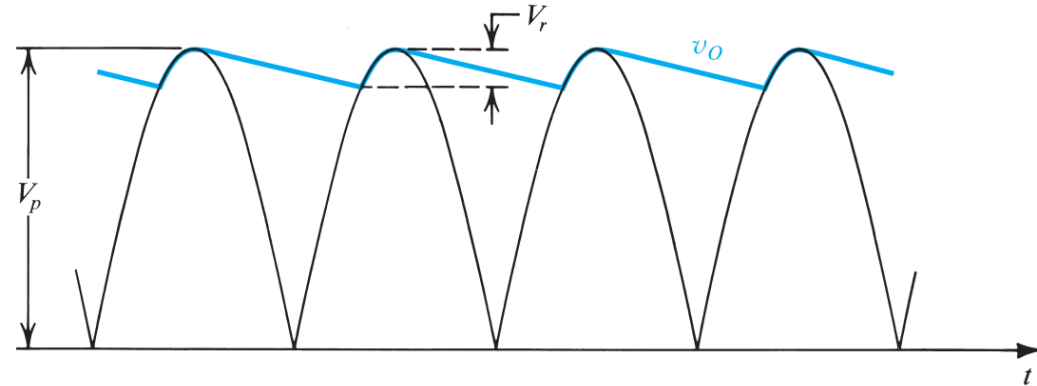
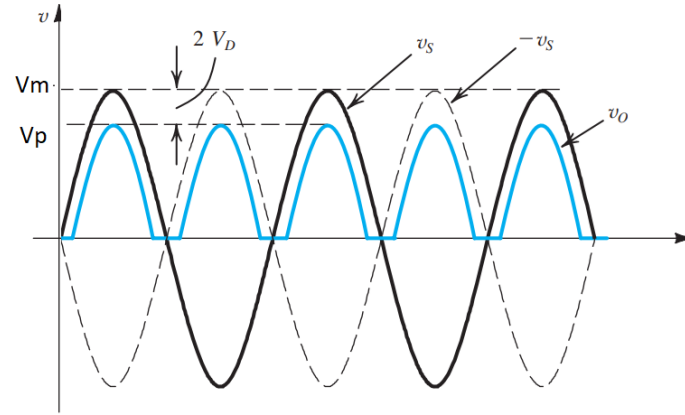
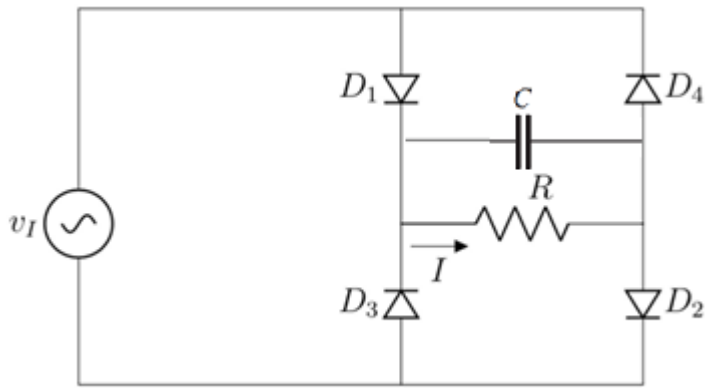


**Full-wave**

# Filtering: Half-wave rectifier



# Filtering: Full-wave rectifier



## Without capacitor

| Rectifier | i/p peak | o/p peak | average                                             |
|-----------|----------|----------|-----------------------------------------------------|
| H/W       | $V_M$    | $V_P$    | $V_{avg}=V_{DC}=\frac{1}{\pi}V_M-\frac{1}{2}V_{Do}$ |
| F/W       | $V_M$    | $V_P$    | $V_{avg}=V_{DC}=\frac{2}{\pi}V_M-2V_{Do}$           |

## With capacitor

| Rectifier | i/p peak | o/p peak          | frequency  | Ripple voltage            | average                             |
|-----------|----------|-------------------|------------|---------------------------|-------------------------------------|
| H/W       | $V_M$    | $V_P=V_M-V_{Do}$  | $f_r=f_i$  | $V_r=\frac{V_P}{f_r R C}$ | $V_{avg}=V_{DC}=V_P-\frac{1}{2}V_r$ |
| F/W       | $V_M$    | $V_P=V_M-2V_{Do}$ | $f_r=2f_i$ | $V_r=\frac{V_P}{f_r R C}$ | $V_{avg}=V_{DC}=V_P-\frac{1}{2}V_r$ |

$$I_{o,avg}=V_{o,avg}/R, V_{rms}=V_p/\sqrt{2}$$



# Example-1

A voltage waveform  $v_i = 10 \sin(100\pi t)$  V is input to a full-wave rectifier with a load resistance of  $R = 50 \text{ k}\Omega$ . Silicon diodes are used in this circuit for which the forward drop is  $V_{D_0} = 0.7 \text{ V}$ .

- (a) **Show** the circuit of the rectifier. **Label** the input and output voltages properly. [2]
- (b) **Calculate** the DC value of the output voltage. [1]
- (c) **Contrast** the value found in part (b) with that when a  $5 \mu\text{F}$  capacitor is connected in parallel with the load. [2]
- (d) **Identify** the two diodes will be ON in the positive half cycle. [1]

Now the two diodes from part (d) are replaced with Germanium diodes [ $V_{D_0} = 0.2 \text{ V}$ ].

- (e) **Explain** the change in the voltage transfer characteristics and output voltage waveform of the circuit. Hence, calculate the peak of the output voltage in this case. [3+1]

## Example-2

The input of a half-wave rectifier is a sinusoidal voltage with peak  $V_M = 10$  V and frequency 60 Hz, and output load resistance is  $R = 2$  k $\Omega$ . Silicon diodes are used in this circuit for which the forward drop is  $V_{D_0} = 0.7$  V.

- (a) Briefly **explain** the purpose of a rectifier and **describe** its operation. [2]
- (b) **Show** the input and output waveforms. [2]
- (c) **Calculate** the DC value of the output voltage. [1]

Now after connecting a capacitor in parallel with the load, the output becomes a ripple voltage  $V_{\text{out}} = V_{\text{DC}} \pm 0.2$  V.

- (d) **Calculate** the peak-to-peak ripple voltage, and from that, the value of the capacitor. [1+2]
- (e) **Calculate** the average of the output voltage  $V_{\text{DC}}$  after connecting the capacitor. [2]

## Example-3

A voltage waveform  $v_i = 5 \sin(200\pi t)$  V is fed into a Full-wave rectifier with a load resistor,  $R = 5 \text{ k}\Omega$ . Silicon diodes are used in this circuit where,  $V_{D_0} = 0.6 \text{ V}$ .

- (a) **Draw** the rectifier circuit. **Label** the input and output voltages properly. Briefly **explain** the application of the circuit. [1+1+1]
- (b) **Calculate** the DC value of the output voltage,  $V_{dc}$  and the output frequency,  $f_o$ . [1+1]
- (c) **Draw** the Voltage Transfer Characteristics (VTC) of the Full-wave rectifier and **label** it properly. [2]
- (d) Now, you have to connect a capacitor in parallel with the load resistor. You have two capacitors of  $5 \mu\text{F}$  and  $1 \mu\text{F}$  at your disposal. Which capacitor will you use? **Explain** briefly with necessary calculations. [3]
- (e) [**Bonus**] A different input waveform is fed into the Full-wave rectifier. The new peak-to-peak ripple voltage is 50% of the previous one calculated from (d) with the  $5 \mu\text{F}$  capacitor. The new output frequency is 300 Hz. **Determine** the equation of the input waveform. [2]

## Example-4

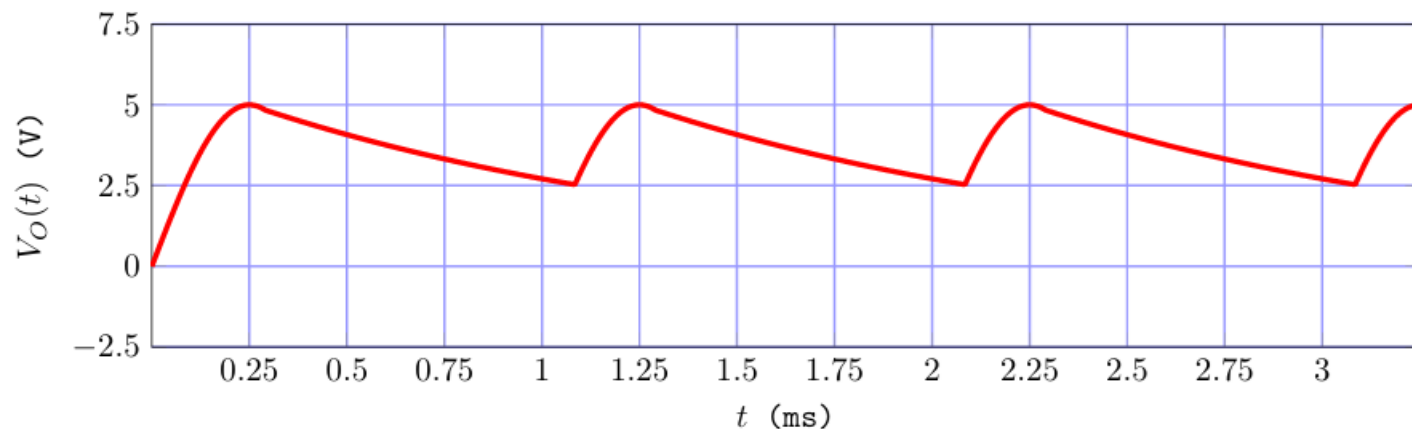


Figure-3

- (c) [2 marks] In *Figure-3*, you are given the output voltage waveform of an unknown rectifier circuit with an output load resistance,  $R = 5k\Omega$ , input frequency,  $f_{in} = 1kHz$ , and  $V_{D0} = 0.5V$ . **Analyze** the waveform in *Figure-3*, and **determine** the output voltage frequency,  $f_{out}$  and **draw** the rectifier circuit with proper labels.
- (d) [2 marks] **Analyze** the circuit in *Figure-4*, and **draw** the output voltage waveform for  $V_{in} = 5\sin(100\pi t)$  V.

## Example-4

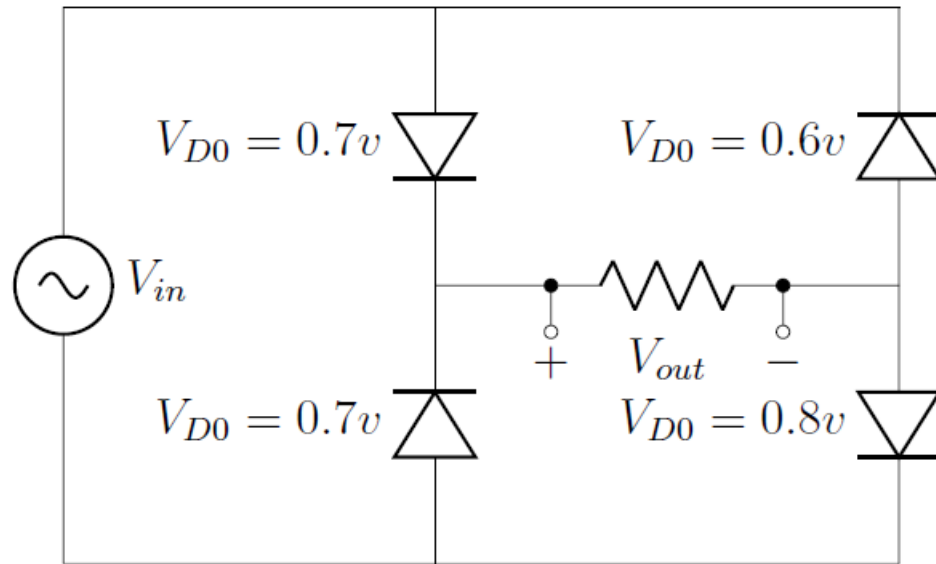
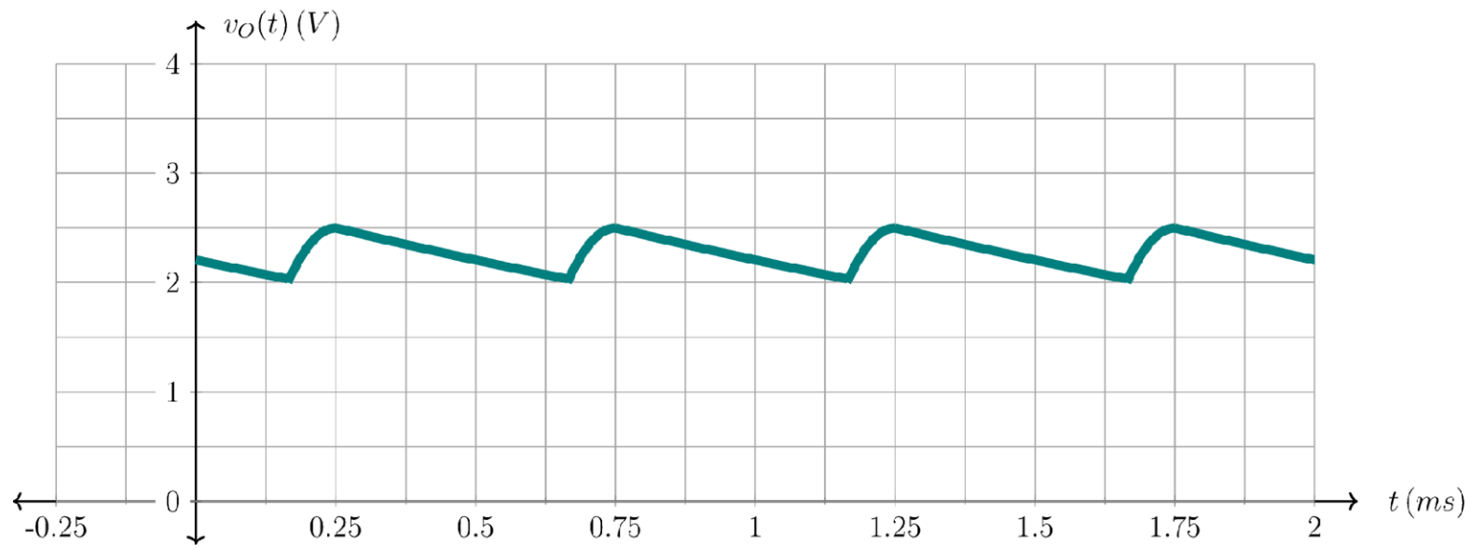


Figure-4

# Example-5

- The following is the rectified output of an ac sinusoidal voltage at  $1\text{ kHz}$  frequency, provided to a load of  $5\text{ k}\Omega$ . The diode(s) used in the rectifier had a cut-in voltage of  $0.5\text{ V}$ .
  - What type of rectifier was utilized?
  - What capacitance was used in the rectifier's design?
  - Write an expression as a function of time for the input voltage?
  - Draw the circuit diagram of the rectifier.



Ans: II.  $0.5\text{ }\mu\text{F}$ ; III.  $\frac{7}{2}\sin(2000\pi t)\text{ V}$

# Thank you